Title and Product: 10F Pusher	**Number**	BER 034
	Revision	Ed02

Revision	Date	Description
01	03/05/2022	Creation of the BER (previous report was BE16 Ed08) Including modifications F611, F602, F632
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INTRODUCTION: PURPOSE AND SCOPE

The purpose of this report is to evaluate the overall biological safety of the following references:

TABLE 1: REFERENCES

Reference	Designation
F3036	10F Pusher

The 10F Pusher is a posing accessory for the VenaTech® Convertible Vena Cava Filter. The VenaTech® Convertible vena cava filter system is intended for percutaneous placement via the femoral or jugular vein for IVC filtration to prevent Pulmonary Embolism (PE).

4435140: VenaTech® Convertible Vena cava filter system
5010028: VenaTech® Convertible Vena cava filter system US

This document is presented in 2 parts :

- A Biological Evaluation Plan which includes the description of the device, its raw materials and manufacturing process, the categorization of the device and the potential biological risks associated, the physical and chemical characterization of the device and the biological test program.
- A Biological Evaluation Report which includes the reports of all biological tests conducted and the overall analysis of the results which allows to conclude on the toxicological risk associated to the device

It should be noted that:

- This biological evaluation report (BER) focuses only on **10F Pusher** described in this report, and does not address the biological risks associated with other devices or accessories.
- This BER is not intended to evaluate the sterility of the device or of its packaging. The validation of the sterilization process and compliance with ISO 10993-7 *"Biological Evaluation of Medical devices – Part 7: Ethylene oxide sterilization residuals"* is documented in the kits associated Sterilization Evaluation Report. However, the influence of sterilization and packaging on the biocompatibility is assessed within the risk analysis of the manufacturing process.

PART 1 : BIOLOGICAL EVALUATION PLAN

1. Standards and references applied for the medical device related to biological evaluation

The list of the applicable documents is presented in Table 2.

TABLE 2 : APPLICABLE DOCUMENTS

Reference	Title
EU MDD	Applicable sections of the European Medical Device Directive 93/42/EEC amended 2007/47/EC
EU MDR	Applicable sections of the European Medical Device Regulation 2017/745/EU (published in Official Journal of the European Union on May 5th, 2017) (GSPR 1, 6, 10.1, 10.2, 10.4)
GMED Technical Memo	Biological evaluation report Information to be provided for assessment
GMED Guide	Biological assessment of medical devices according to the ISO 10993-1 standard
ISO 10993-1 (2018)	Biological evaluation of medical devices – Part 1: Evaluation and testing within a risk management process
ISO 10993-2 (2022)	Biological Evaluation of Medical devices – Part 2: Animal welfare requirements
ISO 10993-3 (2014)	Biological Evaluation of Medical devices – Part 3: Tests for genotoxicity, carcinogenicity and reproductive toxicity
ISO 10993-4 (2017)	Biological evaluation of medical devices — Part 4: Selection of tests for interactions with blood
ISO 10993-5 (2009)	Biological Evaluation of Medical devices – Part 5: Tests for *in vitro* cytotoxicity
NF EN ISO 10993-7 (2008) /A1(2022)	Biological Evaluation of Medical devices – Part 7: Ethylene oxide sterilization residuals
ISO 10993-9 (2019)	Biological Evaluation of Medical devices – Part 9: Framework for identification and quantification of potential degradation products
ISO 10993-10 (2020)	Biological Evaluation of Medical devices – Part 10: Tests for skin sensitization
ISO 10993-11 (2017)	Biological Evaluation of Medical devices – Part 11: Tests for systemic toxicity
ISO 10993-12 (2021)	Biological Evaluation of Medical devices – Part 12: Sample preparation and reference materials
ISO 10993-13 (2021)	Biological Evaluation of Medical devices – Part 13: Identification and quantification of degradation products from polymeric medical devices
ISO 10993-16 (2017)	Biological Evaluation of Medical devices – Part 16: Toxicokinetic study design for degradation products and leachables
ISO 10993-17 (2002)	Biological Evaluation of Medical Devices – Part 17: Methods for the establishment of allowable limits for leachable substances
ISO 10993-18 (2020)	Biological Evaluation of Medical devices – Part 18: Chemical characterization of materials
ISO/TS 10993-19 (2020)	Biological evaluation of medical devices – Part 19: Physico-chemical, morphological and topographical characterization of materials
ISO/TS 10993-20 (2006)	Biological evaluation of medical devices – Part 20: Principles and methods for immunotoxicology testing of medical devices
ISO 10993-23 (2021)	Biological evaluation of medical devices – Part 23: Tests for irritation
ASTM F756-17 (2017)	Standard practice for assessment of hemolytic properties of materials
ISO 14971 (2019)	Medical Devices – Application of risk management to medical devices, Annex I: Guidance on Risk Analysis Procedures for Biological Hazards
ISO/IEC 17025 (2017)	General requirements for the competence of testing and calibration laboratories

2. Formulation, description, manufacturing and use of the medical device

The **10F Pusher** is a posing accessory for the VenaTech® Convertible vena cava filter. It is used to push the filter out of the syringe and into the introducer sheath.



Figure 1: 10F Pusher

2.1 Formulation

The description of the raw materials used to manufacture the **10F Pusher** are provided in the following Table 3:

TABLE 3: DESCRIPTION OF RAW MATERIALS OF 10F PUSHER

Component	Component Manufacturer	Raw Material	Amount (%)	Raw Material Supplier	Type of Patient Contact
Overmolded Hub (F1051)	B. Braun Chasseneuil	STARPYLEN MB008-WT00485	100	EUROSTAR Engineering Plastics ZI	No contact
Tubing (Tube - F1743)	Zeus industrials products LTD	FEP 100 X	100	DuPont de Nemours International S.A.	Ext. Com. Indirect, Blood Path Limited (≤24h)
Ring (Bague de repère - F1461)	SAS Le Guellec	Stainless steel 304L (DIN 1.4306)	100	Salzgitter Mannesmann Stainless Tubes Italia Srl Schoeller Bleckmann Edelstahlrohr GmbH	No contact

The devices must be free from excessive sinks, burns, other surface defects and extrusion defects (nibs, bumps, and ridges).

2.2 Primary packaging

The **10F Pusher** are placed on a support plate before being packaged in a double peelable sheath.

The primary packaging materials are shown in Table 4.

TABLE 4 : PACKAGING MATERIALS USED

Component	Material	Supplier
Support plate (F1888)	Polypropylene PRIPLAK OPALINE bleu 417	Cartolux – Thiers SAS
Internal Peelable sheath 125x1000mm (F1866)	Polyester-based Laminate 35709-G/Tyvek 1073B enduction CR27	Bemis Healthcare Packaging Limited (Nelipak)
External Peelable sheath 150x1120mm (F1876)	Polyester-based Laminate 35709-G/Tyvek 1073B enduction CR27	Bemis Healthcare Packaging Limited (Nelipak)

2.3 Description of the expected and intended biological effect, if applicable

The raw materials used are not intended to have biological effect when implanted in the human body.

No user contact is anticipated as users are supposed to wear gloves during surgery.

2.4 Manufacturing of the medical device

A high level summary of the manufacturing steps performed and associated processing aids is presented in the table below:

TABLE 5: SUMMARY OF MANUFATURING PROCESS

Process designation	Manufacturer	Manufacturing step	Processing aid
Tube manufacturing	Zeus industrials products LTD	Extrusion	None
	B.Braun Chasseneuil	Cleaning	Ethanol 70%
Ring manufacturing[a]	SAS Le Guellec	Machining	Aldraw J-1 Dark
			VICAFIL TFG 4295
			Triboflash 01 (A)
			Carbure de silicium
			Quakercool 7200 HBFF
			Supracut EVO 15M
		Degreasing	Neutralene 2015
			Dowper MC Perchloroethylene Solvent
	B.Braun Chasseneuil	Cleaning	TOPKLEAN MC20D
			neodisher MediClean forte
Pusher manufacturing	B. Braun Chasseneuil	Cutting and tip forming	N/A
		Hub overmolding	N/A
		Ring crimping	N/A
		Cleaning	Ethanol 70%
Support plate manufauring	Cartolux – Thiers SAS	Cutting	N/A
		Cleaning	Ethanol
Peelable sheath Manufacturing	Nelipak	Sealing	N/A
		Cutting	N/A
		Inspection	N/A
	B.BRAUN Chasseneuil	Inspection	N/A
Primary Packaging	B.BRAUN Chasseneuil	Components packaging	N/A
		Lid sealing	N/A
		Breather bag sealing	N/A
Packaging for sterilization	B.BRAUN Chasseneuil	Intermediate packaging	N/A
Sterilization	Sterigenics SAS	Ethylene Oxide Sterilization	ETHYLENE OXIDE

*preconnected access ports only

The primary packaging of the **10F Pusher** is performed by B. Braun Medical (Production site: Chasseneuil du Poitou, France).

The packed **VenaTech® Convertible vena cava filter** including the **10F Pusher** is sterilized using ethylene oxide (EO) by Sterigenics SAS (Rantigny).

The sterilized device is finally sent back to B. Braun Medical (Production site: Chasseneuil du Poitou) where the secondary packaging of the **VenaTech® Convertible vena cava filter** is performed.

2.5 Use of the medical device

Target Population: The 10F Pusher is intended to be used in adults.

[a] See Modification F649 in section 7.2

Claimed clinical performance: The pusher is used to push the filter out of the syringe through the introducer sheath for deployment in the inferior vena cava at the infrarenal position (below the renal veins).

Life duration: The 10F Pusher is for single use only. It is not reusable nor designed to be re-used.

Shelf life: The shelf life of the sterile finished device kits including the 10F Pusher is 5 years.

Storage condition: The storage conditions of the sterile finished device kits including the 10F Pusher are: “Store at room temperature. Avoid freezing and excessive heat”

2.6 Definition of the worst-case

Not Applicable – There is only one device in the scope of this BER

3. Categorization of the medical device: nature and duration of contact

As part of the evaluation process based on the requirements of ISO 10993-1 and 2020 FDA Biocompatibility Guidance, medical devices are categorized based on the type and the duration of patient contact.

When used as intended, the **10F Pusher** is categorized as follow:
- Category: Externally Communicating medical device
- Nature of the contact with the human body: Circulating Blood
- Duration of the contact with the human body: Limited (≤ 24h)

4. Identification of potential biological risks of the medical device / possible biological hazards

Because of the **10F Pusher** categorization, consideration must be given at least to all relevant biological endpoints as described in Table 6.

TABLE 6 : ENDPOINTS TO BE CONSIDERED ON THE 10F PUSHER

Endpoint	Applicable ISO 10993 Standard	Applicable (Y/N)
Physical and/or Chemical Information	ISO 10993-1	Y
Cytotoxicity	ISO 10993-5	Y
Sensitization	ISO 10993-10	Y
Irritation or Intracutaneous Reactivity	ISO 10993-23	Y
Material Mediated Pyrogenicity	ISO 10993-11	Y
Acute Systemic Toxicity	ISO 10993-11	Y
Subacute Toxicity	ISO 10993-11	N
Subchronic Toxicity	ISO 10993-11	N
Chronic Toxicity	ISO 10993-11	N
Implantation Effects	ISO 10993-6	N
Hemocompatibility	ISO 10993-4	Y
Genotoxicity	ISO 10993-3	N
Carcinogenicity	ISO 10993-3	N
Reproductive/ Developmental toxicity	ISO 10993-3	N
Immunotoxicity	ISO 10993-20	N
Degradation	ISO 10993-9, ISO 10993-13, ISO 10993-15	N

(Y) Endpoints to be considered and documented *per* the ISO 10993-1 standard and the 2020 FDA Biocompatibility Guidance
(N) Not applicable

It is important to note that while all of the endpoints listed above must be considered, they can be addressed in a number of different ways including chemical testing, biological testing, and written justification or risk assessment, if relevant information is available.

5. Physical and chemical information for biological risk analysis / medical device characterization

5.1 Material characterization protocol

Materials without any body contact do not have to be considered according to ISO 10993-1 and therefore they are not part of the material characterization.

Evaluation of the chemical nature of the material can take the form of experimental data and/or information on the chemistry of the materials/components involved. Literature studies conducted on the materials help evaluate the biological response and are useful in assessing a finished medical device for its intended use/intended purpose. Some factors that affect the biocompatibility of the material include the identity, concentration, availability, and toxicity of all constituents such as additives, processing aids, and monomers.

Devices must be designed and manufactured in such a way as to reduce to a minimum the risks posed by substances leaching from the product. Special attention shall be given to substances having Carcinogenic, Mutagenic or Reprotoxic (CMR substances) or endocrine-disrupting properties according to European Regulations[b, c].

Identification of the materials and processes used in the manufacture of a medical device enables the intrinsic toxicity of the device to be investigated. The biological risks arising from the use of medical devices are related to two main sources, regardless of the type and duration of body contact:

- Raw materials used to fabricate the medical device, which can combine different chemicals capable of leaching from the device and/or producing degradation products that cause adverse effects once in contact with tissues and body fluids. Moreover, the surface state, porosity, geometry, aspect (solid, gel, liquid) may also influence the biological reaction in the immediate vicinity of the medical device (tissues, bone, blood), i.e., its local tolerance, hemocompatibility;
- Manufacturing process, including machining, assembly, cleaning, packaging and sterilization, are all steps that can provide residues which may be bioavailable once in contact with the body and body fluids and can be responsible for additional biological hazards.

However, in order to estimate the risk of this occurring, it is necessary to obtain information demonstrating to what extent these constituents are available during intended use. The amounts can be estimated from extraction tests on the material.

A chemical characterization to evaluate any risks will request to determine dose-based threshold for these

[b] CMR substances according to Table 3.1 of Annex VI of Regulation (EC) 1272/2008 (Version 2022-12-17, consolidated text): https://eur-lex.europa.eu/eli/reg/2008/1272/2022-12-17

[c] ED Substances according to current ED assessment list and SVHC list from ECHA (European Chemicals Agency): https://www.echa.europa.eu/ed-assessment ; https://www.echa.europa.eu/candidate-list-table

specific chemicals, with demonstrated ability from the laboratory in charge of the analysis to detect these specific substances at, or below, the chemical-specific Analytical Evaluation Threshold.

The ISO 10993 series is intended for use by *professionals, appropriately qualified by training and experience, who are able to interpret its requirements and judge the outcome of the evaluation for each medical device, taking into consideration all the factors relevant to the medical device, its intended use and the current knowledge of the medical device provided by review of the scientific literature and previous clinical experience.*

5.2 Characterization of the materials

5.2.1 Risk analysis of Patient-contacting Raw Materials

As presented in Section 2.1, the raw materials in contact (direct or indirect) with the patient are the following:

TABLE 7: RAW MATERIALS INFORMATION

Raw Material	Raw Material Supplier (Name and Address)	Specifications: MSDS and/or TDS	CAS N°	CMR? ED? SVHC?
FEP 100 X	DuPont de Nemours International S.A. European Technical Centre 146, route du Nant d'Avril-Meyrin-Genève CH-1218 Le Grand Saconnex Switzerland	3_KE[1] 3_KE(DL) [2]	7727-43-7	Not listed

Information provided by the raw material supplier and general information issued from the litterature review are presented in the sections below.

5.2.1.1 Data from suppliers

FEP 100-X

The tubing, supplied by Zeus industrials products LTD, is made of FEP 100-X® provided by DuPont de Nemours International S.A mixed with 9% Barium Sulfate.

FEP 100-X is a melt-processible copolymer of tetrafluoroethylene and hexafluoropropylene, without additives. FEP 100-X does not contains PFOA (perfluorooctanic acid)[1]. According to the declaration of the supplier (3_KE(DL)[2]), it does not contain CMR restricted substances defined in Part 3 of Annex VI Regulation (EC) No. 1272/2008 “Carcinogen category 1A & 1B, Mutagen category 1A & 1B, Reproductive toxicant 1A & 1B” effective May 1, 2020 and Article 59 of Regulation (EC) No. 1907/2006 REACH dated January 16, 2020.

5.2.1.2 Data from literature

General information issued from the literature are summarized in the following Table 8:

TABLE 8: SUMMARY OF LITERATURE REVIEW

Raw Material	Protocol / Report number	Conclusion of the review
FEP	PRO-08-2022[3] REP-08-2022[4]	Fluorinated ethylene polypropylene belongs to the class of fluoropolymers (FPs). As per the review by Henry et al., there are at least four FPs (PTFE, ETFE, FEP, and PFA) that have been demonstrated to meet the OECD’s criteria for Polymers of Low Concern

		(PLCs). As per OECD's definition, "PLCs are polymers deemed to have insignificant environmental and human health impacts. Therefore, these polymers should face reduced regulatory requirements". The evaluation of the most representative FPs by Henry et al. shows that FEP satisfies the widely accepted assessment criteria to be considered PLCs. FEP is a high molecular weight polymer, stable against hydrolysis, light, oxidation, and biodegradation, and thermally stable in the range of 150 °C to 260 °C. Also, it is practically or completely insoluble in water and not soluble in octanol. As solubility in octanol is predictive of lipid solubility, FEP cannot dissolve in cell membrane lipids to gain access to cellular contents. FEP is very similar in composition to PTFE (Polytetrafluoroethylene). FEP shares PTFE's useful properties of low friction and non-reactivity, but are more easily formable. In the review by Henry et al., considering PTFE as a representative substance within the FPs family, the results of irritation and skin sensitization tests (according to ISO 10993-10 standard) on three different physical samples of PTFE substance (patch, fibre, and tube) were non-sensitizing and non-irritating. Taking these results into account, it can be extrapolated that FEP would not cause any local toxic effects. Hexafluoropropylene/Tetrafluoroethylene Copolymer is included on the list of perfluorocarbon resins that the US FDA has determined may be safely used as articles or components of articles intended to contact food, subject to the provisions that are stated in the CFR (21CFR177.1550).37 The EU food contact regulation requires that monomers, other raw materials, and additives used to manufacture food contact polymers should be risk assessed and authorized, as described in Regulation (EU) No 10/2011 on plastic materials and articles intended to come into contact with food and should adhere to the good manufacturing practice Regulation (2023/2006). This regulation lists authorized substances that are permitted to become in contact with food and it also sets their specific migration limit, which is the maximum permitted amount of substance in food that has been determined not to pose a risk to human health, specifically for individual chemicals (e.g., monomer). The monomers, other starting substances, and additives used to manufacture FPs (e.g., PTFE, FEP, and PFA) have been authorized for food contact uses. TFE is included in the positive list of monomers and other starting substances for plastic materials and articles intended to come in contact with food stuffs. A specific migration limit of 0.5 mg/kg (foodstuffs) has been assigned As a representative for FEP, medical devices containing PTFE have been evaluated using ISO 10993 and US Pharmacopeia (USP) Class VI standards and have been determined to be biocompatible in their intended uses, over the years. The clinical history of the safe implantation of more than 40 million PTFE medical devices over 40 years, extensive toxicity data, preclinical data, and chemical extractables and migration testing confirmed that fluoropolymers are not bioavailable. Thus, concluding that fluoropolymers are safe for these intended uses and support the conclusion that fluoropolymers including FEP should be considered PLCs Based on the available literature, FEP used as a coating on metallic stents has proved to be safe in patients. The available data for FEP copolymers, TFE and HFP, is based on animal inhalation exposure where both showed some carcinogenic effects in animal studies, however did not show any genotoxic effects. With respect to human exposure characterisation (and subsequent hazard assessment), if LMW components i.e. monomers like TFE and HFP are unable to migrate from the polymer matrix to any relevant extent, the oral and dermal routes of exposure or systemic toxicity upon inhalation are unlikely to be of relevance for these compounds. In cases where minor components of polymer products are assumed to become physically available, the hazard assessment should be conducted on the components as such and considered for further safety evaluation. Also, considering the minimal availability of data or information on FEP, further evaluation on the safety/biocompatibility aspect can be considered based on the use of the raw material in the medical device in view of its intended application/use, contact type and duration.
Barium Sulfate	PRO-17-2022[5] REP-17-2022[6]	Literature on safety/toxicity of barium sulfate is minimal. However, based on the available literature information, barium sulfate is known to have low or no toxicity because of its insoluble nature. The insoluble, nontoxic nature of barium sulfate has made it practical to use this particular barium compound in medical applications as a contrast media and is frequently used as a benign, radiopaque aid to x-ray diagnosis in colorectal and some upper gastrointestinal examinations, because it is normally not absorbed by the body after oral intake. Under these routine medical situations, barium

		sulfate is generally safe. However, barium sulfate may potentially be toxic when it is introduced into the gastrointestinal tract under conditions where there is colon cancer or perforations of the gastrointestinal tract and barium is able to enter the blood stream.4 Based on the available literature search and the data available for toxicity endpoints like acute, repeated dose toxicity, irritation, sensitization, genotoxicity, carcinogenicity, reproductive and development toxicity, for barium sulfate or its structural analogues it can be concluded that barium sulfate has a low toxicity potential with no adverse effects and can be safely used in medical applications. Also, barium sulfate with dermal fibroblast cells showed to be noncytotoxic. Barium sulfate (BaSO4) was the first radiopaque material to be widely compounded in medical formulations and is the most common filler used with medical-grade polymers and has a long history of safe use, thus concluding its safety.

5.2.1.3 Summary of risk analysis of Patient-contacting Raw Materials

The collected data from the suppliers, as well as the data obtained from literature search, allows to identify that some of them are identified as of potential concerns. These concerns are either from the raw materials themselves, or from the substances that could be hazardous if they are released into the patients.

For the list of concerns provided below, the hazards (only) are being identified The **10F Pusher** tubing are made of fluoroethylene propylene including Barium sulfate. Considering the literature above and the limited contact of this device with the patient, no health concerns are expected from the use of the **10F Pusher**.

5.2.2 Risk analysis of Manufacturing Process

5.2.2.1 Processing aids

CAS numbers of the processing aids are used to determine whether there are any classifications relating to human health according to the European Medical Device Regulation 2017/745/EU or the EC regulation 1272/2008 as amended.

When possible, the processing aids have been reviewed for their Summary of Classification and Labelling (CLP) status, either as reported in the Safety Data Sheets (SDS) or in the European Regulation (EC) No. 1272/2008 (CLP Regulation) as amended.[b] The noted properties of concern were irritation, sensitization, mutagenicity, carcinogenicity, and reproductive toxicity (including lactation effects).

The Endocrine Disrupting (ED) properties of each chemical were also assessed based on information, if any, detailed within the REACH *Registry of SVHC intentions* or *endocrine disruptor assessment*.[c]

The summary of Health Hazard Statement Codes and ED assessment on the processing aids used during the manufacture of **10F Pusher** is summarized in Table 9.

TABLE 9 : PROCESSING AIDS – SUMMARY OF HEALTH HAZARD STATEMENT CODES AND ENDOCRINE DISRUPTION

Processing Aids	Composition	CAS Number	Health Hazard Statement Code*	CMR? ED? SVHC?
ALCOOL ETHYLIQUE 70% (PA0002)[7]	Mixture	N/A	H319	N/A
	Distilled water	7732-18-5	N/A	N/A
	Ethanol	64-17-5	H319	N/A

(n/a): chemical is not listed in the queried documents.
(*): Health hazard statement code as mentioned in the SDS
Health hazard codes:
H**319**: Causes serious eye irritation ;

Based on information presented in Table 9, no processing aids as prepared and used in the manufacture of the device contain substances classified CMR 1A or 1B or with endocrine-disrupting properties.

Some processing agents are classified as harmful or toxic. Toxicity for eye contact, is considered to be non-clinically relevant since the final product is not in contact with the eyes.

As the **10F Pusher** are commercialized, the manufacturing process and cleaning is validated to remove or minimize the amount of unwanted residues of manufacturing. Therefore, the patient would unlikely be exposed to these processing aids at level of toxicological concerns.

To assess the biological risks related to potential processing aids residues, any chemical characterization or biological test are conducted on a finished device which has followed all manufacturing process steps.

5.2.2.2 Primary Packaging

The sterility of the **10F Pusher** is maintained by the packaging until its point-of-use.

Any contaminants present on the surface of the primary packaging may migrate to the medical device, essentially to the outer surface of the medical device itself. Based on this, consideration must also be given to the primary packaging.

The primary packaging materials are provided by well-known suppliers of packaging materials for medical devices. It is reasonable to consider that the selected materials are good materials to reduce individual biological risk by design as recommended by EN ISO 14971 principles.

In order to assess the biological risks related to the primary packaging, any chemical characterization or biological test are conducted on a finished device packaged in its primary packaging.

5.2.2.3 Sterilization

The packed **VenaTech® Convertible vena cava filter** including the **10F Pusher** is sterilized using ethylene oxide (EO) by Sterigenics SAS (Rantigny).

EO sterilization is known to produce highly reactive residues of ethylene oxide, CAS No. 75-21-8, classified and labeled with CMR health hazard codes H350 (Carc. 1B), H340 (Muta. 1B) and H360Fd (Repr. 1B) *per* the 14th ATP of the EU Regulation (EC) No. 1272/2008.

Compliance with ISO 10993-7, including evaluation of sterilization-related residues present on the final, finished device, is documented in the associated Sterilization Evaluation Report. Ethylene sterilization processes are validated and the ethylene oxide sterilization residuals meet the ISO 10993-7 requirements.

It is considered herein that the biocompatibility of the devices may be affected by the sterilization process; therefore, only testing of a final, finished (*i.e.*, sterile) device can confirm that the sterilization process does not affect the biocompatibility of the device.

Furthermore, the ethylene oxide sterilization process requires heat and humidity that can increase the migration of substances from the primary packaging to the surface of the device.

Therefore, taking into account that the finished device can be re-sterilized if necessary, all chemical and biological tests are conducted on finished devices sterilized twice by ethylene oxide.

5.2.2.4 Summary of risk analysis of Manufacturing process

After an analysis of the hazard identification of the manufacturing processes used to construct the **10F Pusher**, it appears that no processing aids are associated with classifications of greatest toxicological concern (SVHC, CMR 1A or 1B substances).

The primary packaging of the **VenaTech® Convertible vena cava filter** including the **10F Pusher** is made of support plate in a double peelable sheath. These raw materials are usual in primary packaging of medical devices, they are well characterized with a long history of clinical use in approved and marketed medical devices packaging.

Sterilization by Ethylene Oxide is known to produce highly active residues. The **10F Pusher** are sterilized with validated ethylene oxide sterilization processes and the ethylene oxide sterilization residuals meet the ISO 10993-7 requirements.

While the manufacturing process is expected to minimize, and eliminate as much as possible, the quantity of unwanted residues, the combined residual risks as defined by ISO 14971 cannot be addressed by a literature review since manufacturing residues are specific and unique to each manufacturer having their own manufacturing processes.

Furthermore, the ethylene oxide sterilization process requires heat and humidity that can increase the migration of substances from the primary packaging to the surface of the device.

Accordingly, in order to adress all risks related to the manufacturing process, primary packaging and sterilization process, all chemical and biological tests are performed on a final finished device, having followed all manufacturing steps, packaged in its primary packaging and sterilization twice by ethylene oxide.

5.2.3 Conclusion on the need of chemical characterisation test

The **10F Pusher** should be subjected to chemical analysis following the appropriate manufacturing and sterilization processes, encompassing a variety of chemical characterization tests to evaluate the amount, identity, and characteristics of the extractables using methods prescribed in ISO 10993-18: Chemical Characterization of Materials.

As such, the following programs of testing is recommended for the **10F Pusher**:

- Exagerated extractions in polar, and non-polar vehicles for determination of total Non-Volatile Residue (NVR).
- Fourier Transform Infrared (FTIR) spectroscopy of residues obtained from the test article extracts.
- Determination of trace metals/elements using Inductively Coupled Plasma - Mass Spectrometry / Optical Emission Spectroscopy (ICP-MS) (polar extract only).
- Gas Chromatography with Mass Spectrometry (GC-MS) for detection of Semi-Volatile Organic Compounds (SVOCs) from the test article extracts.

- Ultra-Performance Liquid Chromatography with Mass Spectrometry (UPLC-MS) for detection of Non-Volatile Organic Compounds (NVOCs) from the test article extracts.
- Headspace GC-MS for detection of Volatile Organic Compounds (VOCs).

Once obtained, the chemical characterization data should be subjected to toxicological evaluation per ISO 10993-17 and ISO/TS 21726 to determine if there are unmitigated risks associated with patient exposure to extractable chemicals during clinical use. These data serve a number of useful purposes in the toxicological evaluation of medical devices including:

- The chemical characterization data obtained will allow for extractable chemicals to be identified, in some cases quantified, and toxicologically evaluated. These data will enable a more thorough evaluation to be performed with less leveraged data.
- Chemical characterization will provide the baseline data set useful for supporting the equivalence assessments with other devices.
- Chemical characterization and toxicological profile as obtained can be used for future device iterations for purposes of conducting equivalence assessments, and further considered not to repeat biological testing conducted.

5.3 Chemical characterization test

The chemical characterization study was performed on the 7F Dilator (F3026) to identify and quantitate the extractables and/or leachables that may be released from the test article. This study was conducted based on guidance provided in ISO 10993-18, Biological evaluation of medical devices - Part 18: Chemical characterization of medical device materials within a risk management process. The Test report is registered under number C0236D[8].

An exaggerated extraction using PW and Hex was performed on the test article. The resulting extracts were analyzed by FTIR spectroscopy, GC-MS and UPLC-UV-MS. The PW test article extract was additionally analyzed by HS-GC-MS and ICP-MS/OES and IC.

Based on the data obtained from the chemical characterization testing, a Chemical Risk Assessment **(CRA 003[9]),** was conducted in order to determine if there are extractable substances at levels that could present a toxicological risk.

Overall, the chemical characterization results obtained on the 7F Dilator FEP Brachial F3026 indicate that the likelihood of an adverse patient reaction from exposure to a leachable chemical is minimal and associated toxicological risks are acceptable.

Justification of the selection of the test article

The chemical characterization tests were conducted on a representative device : the 7F Dilator F3026.

The 7F dilator has characteristics relevant to be compared to those of the 10F Pusher.

TABLE 10: COMPARISON BETWEEN PATIENT-CONTACTING PARTS OF 10F PUSHER VS 7F DILATOR

10F Pusher (F3036)	7F Dilator (F3026)	10F Pusher	7F Dilator (F3026)	
Parts in contact with the patient		Raw materials		Comparison

Tubing (F1743)	7F Tubing (F1754, F1746)	FEP 100-X PFOA free (9% BaSO4 included)	FEP 100-X PFOA free (9% BaSO4 included) + Cobalt pigment S911.5 (<0.5%)	Same raw material provided by same supplier No colorant in 10F Pusher Direct contact with circulating blood Smaller size of the device for the 10F Pusher

The 7F Dilator F3026, is sharing the same tubing material with the 10F Pusher, with the addition of a blue dye. The components are supplied by the same manufacturing using the same manufacturing process.

The test article was packaged in its primary packaging which is identical to the 10F Pusher primary packaging.

As the packed finished device can be re-sterilized if necessary, biocompatibility tests are conducted on test articles sterilized twice by ethylene oxide.

Therefore, results obtained on this device are considered as relevant to address the biological safety risks associated with the 10F Pusher.

Relevance of the tests performed

The chemical characterization tests have been conducted in line with the most recent ISO 10993-18 standard.

Evidence of the ISO 17025 accreditation or equivalence of the testing laboratory:

All chemical characterization tests have been performed by laboratories with accreditation to ISO 17025. The list of laboratories and certificate identification number are presented in the following Table 12.

TABLE 11 : LABORATORIES ACCREDITATION

Name and Address of the laboratory	SMQ certificate identification number (name of certifying body / validity date)
NAMSA Industrie Center Obernburg 63784 Obernburg am Main - Germany	Certificate n° D-PL-21154-01-00_2019[10] (DakkS GmbH / 26.09.2022)

5.4 Potential of degradation

Conducting a systematic evaluation of the potential and observed degradation of this medical device is not considered to be relevant through the design and performance of in vitro degradation studies for the following reasons:

- As stipulated in ISO 10993-1 subclause 4.2, in the selection of materials to be used in the medical device manufacture, the first consideration shall be fitness for purpose with regard to characteristics and properties of the material. Accordingly, the materials selected to manufacture components intended to be in limited indirect contact with biological fluids were appropriately selected for their intended use;
- As stipulated in ISO 10993-9 preliminary considerations, it is neither necessary nor practical to conduct degradation studies for all medical devices. According to Annex A of ISO 10993-9, degradation studies are not relevant since :
    - The **10F Pusher** have a limited contact duration (<24 hours)
    - The **10F Pusher** are not designed to be absorbable
    - The materials used to manufacture the **10F Pusher** are well-known materials with an established history of biocompatibility.

6. Biological testing program

6.1 Justification on the need or not to perform biological evaluation tests

The 10F Pusher is a externally communicating medical device in limited contact (<24h) with circulating blood. Based on ISO 10993-1 and considering the 2020 FDA Guidance, the biological risks that need to be addressed are cytotoxicity, sensitization, irritation or intracutaneous reactivity, material mediated pyrogenicity, acute systemic toxicity and hemocompatibility.

From a regulatory standpoint, while all of the endpoints specified above must be considered on the 10F Pusher, they can be addressed in a number of different ways, including testing, written justification or risk assessment if relevant information is already available.

For the raw materials of the **10F Pusher** that contact the patient for a limited duration, the expected risks are limited. Based on the performed analysis, no health concerns are expected from the use of the 10F Pusher.

In the manufacturing process, for processing aids for which information was available, the raw materials are not associated with health concerns. The chemical characterization with toxicological risk assessment was conducted to confirm the absence of risks.

While no concerns are expected related to the primary packaging raw materials, sterilization is conducted with ethylene oxide, which is known to be associated with highly reactive residues.

A comprehensive chemical characterization was conducted on a similar device. Evaluation covered total NVRs with FTIR of the residues after exagerated extractions in three different vehicles, ICP (MS or OES), SVOCs, NVOCs and VOCs detections. Such testing allowed to establish the toxicological profile of a device representative of the **10F Pusher**.

Overall, the chemical characterization results indicate that the likelihood of an adverse patient reaction from exposure to a leachable chemical is minimal and associated toxicological risks are acceptable. However, the lack of information of the risk of combination of the substances detected suggest that biocompatibility tests are needed to confirm the acceptability of the risk.

Considering all the above, biological testing need to be conducted to provide information on the risks associated with with cytotoxicity, irritation, sensitization, pyrogenicity, hemocompatibility, which are information not provided through chemical characterization and toxicological analysis.

The estimation of risks for the **10F Pusher** has been reviewed. Considering all of the above, the proposed risk-control strategy to address the biocompatibility of the **10F Pusher** is detailed in the following section.

6.2 Determination of the testing program

Considering that the biological testing is intended to support the safety of these components, tests shall be carried out according to Good Laboratory Practice (GLP) principles (*per* ISO 10993-1, subclause 6.3.1 and the FDA Biocompatibility Guidance, section IV, B, 4)

- **Cytotoxicity** (ISO 10993-5): Highly sensitive *in vitro* test intended to screen biologically harmful extractables in the absence of protective mechanisms that normally assist cells within the body. A quantitative method such as MTT or MTS is recommended to be conducted. An extraction at 37°C

for 72 hours (instead of 24 hours *per* ISO 10993-5, subclause 4.2.3.2.) is recommended for devices with prolonged or long-term contact duration, as a worst-case.

- **Sensitization** (ISO 10993-10): *In vivo* test intended to determine whether significant quantities of leachables/extractables could induce delayed-type hypersensitivity (Type IV) reactions. The ISO 10993-10 recommends a Guinea Pig Maximization test to performed as it is the most sentitive method and the occlusive system test is recommended for topical products. Two extractions in polar and apolar extraction vehicles (such as 0,9% sodium chloride and sesame oil) must be performed. An extraction at 50°C for 72 hours is recommended as *per* ISO 10993-12.
- **Intracutaneous reaction** (ISO 10993-10: 2010, ISO 10993-23: 2021): *In vivo* test intended to evaluate irritation potential of leachables/extractables after a short-term exposure by the intracutaneous route. For medical devices used as implants, the ISO 10993-23, paragraph 7.3 recommends *in vivo* intracutaneous injection tests which are more relevant. An extraction at 50°C for 72 hours using polar and apolar extraction vehicles (such as 0,9% sodium chloride and sesame oil) is recommended as *per* ISO 10993-12.
- **Acute systemic toxicity** (ISO 10993-11): *In vivo* test intended to estimate, during a period of less than 24 hours, the potential harmful effects of systemic exposure to leachables/extractables. An extraction at 50°C for 72 hours using polar and apolar extraction vehicles (such as 0,9% sodium chloride and sesame oil) is recommended as *per* ISO 10993-12.
- **Material-mediated pyrogenicity** (ISO 10993-11 and Regional Pharmacopoeia): *in vivo* test intended to detect substances inducing a febrile reaction in the patient. Rabbit Pyrogenicity tests following the European and US Pharmacopeia are recommended.
- **Hemocompatibility:**
    - **Hemolysis** (ISO 10993-4 and ASTM F756): This *in vitro* test is intended to evaluate the potential of the test article to cause red blood cell hemolysis. A hemolytic index is determined. This test is recommended to be carried out with direct and indirect contact methods.
    - **Complement activation** (ISO 10993-4): This in vitro test is intended to evaluate the potential of the device to activate the complement system. The test article is incubated in normal human serum (NHS) and SC5b-9 is measured in the exposed serum using an enzyme immunoassay.
    - **Partial Thromboplastin Time** (ISO 10993-4 and ASTM F2382):This in vitro test is intended to to determine the potential to cause an effect on the coagulation cascade via the intrinsic coagulation pathway. The average clotting time of a plasma exposed to the test article is compared to the clotting time of a negative reference material.

6.3 Rationale for tests not completed

N/A – All recommended biological tests are performed.

6.4 Summary of testing program

The testing program is summarized in Table 13.

Title and Product: 10F Pusher	**Number**	BER 034
	Revision	Ed02

TABLE 12: METHODS OF CONSIDERATION OF BIOLOGICAL ENDPOINTS FOR THE 10F PUSHER

Biological Testing program*		**10F Pusher** (contact < 24 hours)
Cytotoxicity		T
Sensitization		T
Intracutaneous reaction / irritation		T
Systemic Toxicity	Material Mediated Pyrogenicity	T
	Acute Systemic Toxicity	T
	Subacute/subchronic (13 weeks) Toxicity	N
	Chronic Toxicity	N
Hemocompatibility	Hemolysis	T
	Complement activation	T
	Partial Thromboplastin Time	T
Local Tissue Effects (Implantation)		N
Genotoxicity	Bacterial Reverse Mutation Assay	N
	Mouse Lymphoma Assay	N
Carcinogenicity		N
Immunotoxicity		N
Reproductive/developmental toxicity		N
Degradation		N

(T): Test recommended to be performed. Biological tests should be conducted according to GLP regulations (FDA GLP).
(N): Not applicable
(*): *Per* ISO 10993-12, it is recommended that extractions be performed at 50°C for 72 hours (apart for cytotoxicity).

PART 2 : BIOLOGICAL EVALUATION REPORT

6.5 Biological tests performed

All of the tests have been conducted in November 2021 to January 2022. A high level summary of the biological tests conducted is presented in the following Table 14.

TABLE 13 : SUMMARY OF BIOLOGICAL TESTS CONDUCTED

Study Number	Standards Applied	Biocompatibility Test	Extraction conditions	Results
Test Article: 7F Dilator FEP Brachial, reference: F3026, batch: DE8902, 2xEO sterile				
C0236A[11]	ISO 10993-5 (2009) ISO 10993-12 (2021)	ISO MTS cytotoxicity test	EMEM10 37°C, 24h, 3cm²/mL	Pass
C0236B[12]	ISO 10993-10 (2010) ISO 10993-12 (2021)	ISO Intracutaneous Study in Rabbits (Two Extracts)	SC, SO 50°C, 72h, 3cm²/mL	Pass
C0236C[13]	ISO 10993-11 (2017) ISO 10993-12 (2021)	ISO Acute Systemic Toxicity Study in Mice (Two Extracts)	SC, SO 50°C, 72h, 3cm²/mL	Pass
C0236E[14]	USP 2021 general chapter 151 ISO 10993-11 (2017) ISO 10993-12 (2021)	Rabbit Pyrogen Study according to United States Pharmacopeia - National Formulary 2021 (USP)	SC 50°C, 72h, 3cm²/mL	Pass
C0236F[15]	Ph Eur 10th, Section 2.6.8 ISO 10993-11 (2017) ISO 10993-12 (2021)	Rabbit Pyrogen Study according to European Pharmacopeia	SC 50°C, 72h, 3cm²/mL	Pass
C0236G[16]	ISO 10993-4 (2017) ISO 10993-12 (2021)	SC5b-9 Complement Activation Assay	NHS 37°C, 1h, 3cm²/mL	Pass
C0236H[17]	ISO 10993-4 (2017) ISO 10993-12 (2021) ASTM-F756 (2017)	ISO 10993 and ASTM Hemolysis	CMF-DPBS 50°C, 72h, 3cm²/mL	Pass
C0236I[18]	ISO 10993-10 (2010) ISO 10993-12 (2021)	ISO Guinea Pig Maximization Sensitization Test (Two Extracts)	SC, SO 50C, 72h, 3cm²/mL	Pass
C0236J[19]	ISO 10993-4 (2017) ASTM F2382 (2018)	ASTM Partial Thromboplastin Time	citrated human plasma 37°C, 15min, 6cm²/mL	Pass

EMEM10: Eagle's minimal essential medium with 10% Bovine fœtal serum; SC: 0,9% Sodium Chloride; SO: Sesame oil; NHS: Normal human serum ; CMF-DPBS: Calcium and Magnesium free Dulbecco's phosphate-buffered saline

Justification of the selection of the test article

The biological tests were conducted on the same test article than the chemical characterization test.

The justification of the selection of the test article is presented in Section 5.3.

Relevance of the tests performed

All but two of these tests have been conducted in line with their most recent ISO 10993 respective parts. Irritation has been conducted per ISO 10993-10 (2010) requirements while now ISO 10993-23 (2021) is

applicable. ISO 10993-23 (2021) is presenting alternative *in vitro* testing to *in vivo* ones for evaluation irritation. However, when implants are considered, *in vivo* tests remains recommended. Sensitization has been conducted per ISO 10993-10:2010 while now ISO 10993-10 (2021) is applicable. ISO 10993-10 (2021) has been updated to reflect the split between the irritation and the sensitization guidances. Therefore, the creation of the ISO 10993-23 and the update of the ISO 10993-10 are not impacting the validity of the results.

Regarding extraction methods, ISO 10993-12:2021 currently applies. Considering that the updates to ISO 10993-12 mainly discuss the conditions for extractions of devices that are intended to degrade, no impacts of the update of this standard are seen on the tests conducted.

Evidence of the ISO 17025 accreditation or equivalence of the testing laboratory:

All biocompatibility tests have been performed by laboratories with accreditation to ISO 17025. The list of laboratories and certificate identification number are presented in the following Table 15.

TABLE 14: LABORATORIES ACCREDITATION

Name and Address of the laboratory	SMQ certificate identification number (name of certifying body / validity date)
NAMSA 115 Chemin de L'Islon 38670 Chasse-sur-Rhône - France	Certificate n°1-0628 rev8[20] (Cofrac / 31.12.2022)

7. Overall analysis of the results

7.1 Summary and analysis of all the data described and/or results obtained in the previous stages

This biological risk assessment is supported by information on the **10F Pusher** materials of construction, available information on the manufacturing process, gathered toxicological data on these materials, and existing data on this device.

Considering all gathered data, this risk assessment indicates that a comprehensive biocompatibility testing program was needed to establish the biocompatibility profile of the **10F Pusher** including chemical characterization and biological testing.

Biological testing provided information on the risks associated with cytotoxicity, irritation, sensitization, pyrogenicity, hemocompatibility, which are information not provided through chemical characterization and toxicological analysis.

Based on the results of the biological testing, the **10F Pusher** are considered non-cytotoxic, non-sensitizing, non-irritant, non-pyrogenic, not hemolytic, and not an acute toxicant.

Finally, the data obtained from the toxicological risk assessment conducted on the substances identified by chemical characterization allow to consider that the risks of systemic toxicity are acceptable.

TABLE 15: SUMMARY OF RISK ASSESSMENT

Endpoint	Toxicological Risk	Risk control
Cytotoxicity	Acceptable	Cytotoxicity test has been performed on a representative product. Test article has shown no signs of cell toxicity at L-929 mouse fibroblasts. The final product meets the requirements of the test and is judged no cytotoxic.
Sensitization	Acceptable	Sensitization test has been performed on a representative product. The topical application of the 0,9% sodium chloride and sesame oil extract of the test article did not induce delayed sensitization in the guinea pig. The test article is not considered a sensitizer in the guinea pig maximization model.
Intracutaneous irritation	Acceptable	Intracutaneous reaction test has been performed on a representative product. The test article met the requirement of the test following intracutaneous injection in rabbits after extraction in 0,9% sodium chloride and sesame oil. The test article is not considered as irritant.
Pyrogenicity	Acceptable	Pyrogenicity tests have been performed on a representative product. The test article met the requirement of the USP and EP. The test article was judged non-pyrogenic.
Acute Systemic Toxicity	Acceptable	Acute systemic toxicity test has been performed on a representative product. Under the conditions of this study, there was no mortality or evidence of systemic toxicity from the 0,9% sodium chloride and sesame oil extracts injected into mice. Each test article extract met the requirements of the study.
Hemocompatibility (Hemolysis)	Acceptable	Hemolysis tests (direct and indirect) have been performed on a representative product. Under the conditions of this study, the mean hemolytic index for the test article by direct contact was of 0.2% and the mean hemolytic index for the test article extract was of 0.1 %. The test article extract was non hemolytic.
Hemocompatibility (Complement)	Acceptable	Complement activation assay test has been perfomed on a representative product. The concentration of SC5b-9 in the test article sample was 6235.0 ± 164.1 ng/mL (mean ±SD) and was not statistically higher than the activated NHS control and was not statistically higher than the negative control. As a result, the test article was not considered to be a potential activator of the complement system.
Hemocompatibility (PTT)	Acceptable	ASTM Partial Thromboplastin Time has been performed on a representative product. Under the condition of the study, The test article average clotting time was lower and was not statistically significant when compared to the negative reference material. The test article average clotting time was lower and was statistically significant when compared to the negative control. The test article met the requirements of the test.

7.2 Reassessment of Risk

This risk assessment is valid for the current iteration of the **10F Pusher**. It applies to devices manufactured using the current processes and techniques.

As specified in the Clause 4.9 of the current ISO 10993-1, the biological risk assessment of **10F Pusher** shall be re-evaluated if any of the following occur:

- any change in the source or in the specification of the materials used in the manufacture of the product,
- any change in the formulation, processing, primary packaging, or sterilization of the product,
- any change in the manufacturer's instructions or expectations concerning storage, e.g. changes in shelf life and/or transport,
- any change in the intended use of the product,
- any evidence that the product can produce adverse biological effects when used in humans.

During the redaction of the document, the following changes was implemented:

- **F649**: Le Guellec, manufacturer of the Ring, removed the processing aid ILOFORM TDN 81 of its manufacturing process because it conained tallow derivatives. This modification has no impact on the

biological evaluation it removes riks of residues on finished products and therefore remove the associated toxicological risks. Furthermore, the ring has no contact with the patient.

The following changes were implemented and described in the previous editions of this Biological Evaluation Report:

- **F602**: Correction of the composition of the tube. The raw materials are not changed, it is juste a correction of the designation. No impact on the biological evaluation.
- **F611**: Correction of the composition of the raw material of the hub according to data from the supplier. No impact on the biological evaluation has there is no change of the raw material.
- **F632**: Change of nomenclature. This modification has no impact on the biological evaluation as only the component number in the specfication are changed.

8. References

Figure 1: 10F Pusher

Table 1: References
Table 2 : Applicable documents
Table 3: Description of raw materials of 10F Pusher
Table 4 : Packaging Materials used
Table 5: Summary of manufaturing process
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Table 12 : Laboratories accreditation
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Table 14 : Summary of biological tests conducted
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Table 16: Summary of risk assessment

Referenced documents:

[1] **3_KE:** Material Safety Data Sheet of FEP 100 X

[2] **3_KE(DL):** Declaration letter on FEP 100 X

[3] **PRO-08-2022**: Literature Search and Review plan for FEP

[4] **REP-08-2022**: Final Report, Literature review for FEP

[5] **PRO-17-2022**: Literature Search and Review plan for Barium Sulfate

[6] **REP-17-2022**: Final Report, Literature review for Barium Sulfate

[7] **PA0002_MSD_Ethanol 70_ZCP211_Alcogroup**: Material Safety Data Sheet of ALCOOL ETHYLIQUE 70%

[8] **C0236D_CHEM**: Chemical Characterization program of 7F Brachial Dilator

[9] **CRA003** : Chemical Risk Assessment related to C0236D

10 **GLP_D-PL-21154-01-00_2019_NAMSA Obernburg**: ISO 17025 certificate N° D-PL-21154-01-00_2019 of NAMSA Obernburg

[11] **C0236A_CYT_MTS:** Biological Test Report - ISO MTS cytotoxicity test - 7F Dilator FEP Brachial (F3026)

[12] **C0236B_IRR_IC:** Biological Test Report - ISO Intracutaneous Study in Rabbits (Two Extracts) - 7F Dilator FEP Brachial (F3026)

[13] **C0236C_TOX_Acute:** Biological Test Report - ISO Acute Systemic Toxicity Study in Mice (Two Extracts) - 7F Dilator FEP Brachial (F3026)

[14] **C0236E_PYR_USP:** Biological Test Report - Rabbit Pyrogen Study according to United States Pharmacopeia - National Formulary 2021 (USP) - 7F Dilator FEP Brachial (F3026)

[15] **C0236F_PYR_EP:** Biological Test Report - Rabbit Pyrogen Study according to European Pharmacopeia - 7F Dilator FEP Brachial (F3026)

[16] **C0236G_HEM_Comp:** Biological Test Report - SC5b-9 Complement Activation Assay - 7F Dilator FEP Brachial (F3026)

[17] **C0236H_HEM_Hemolysis:** Biological Test Report - ISO 10993 and ASTM Hemolysis - 7F Dilator FEP Brachial (F3026)

[18] **C0236I_SEN_GPM:** Biological Test Report - ISO Guinea Pig Maximization Sensitization Test (Two Extracts) - 7F Dilator FEP Brachial (F3026)

[19] **C0236J_HEM_PTT:** Biological Test Report - ASTM Partial Thromboplastin Time - 7F Dilator FEP Brachial (F3026)

20 **GLP_1-0628_rev8_NAMSA Lyon**: ISO 17025 certificate N° 1-0628 rev.8 of NAMSA Chasse-sur-Rhône